

PROJECT REPORT

VOLTEDGE EV ANALYTICS

1. INTRODUCTION

1.1 Project Overview

VoltEdge is a comprehensive data analytics portal designed to bridge the information gap in the Indian Electric Vehicle (EV) market. By integrating a **PostgreSQL** backend with **Tableau** visualizations and a **Flask** web interface, the project provides real-time insights into charging infrastructure, vehicle efficiency, and market pricing.

1.2 Purpose

The purpose of this project is to empower three key stakeholders: city planners (infrastructure optimization), fleet managers (operational efficiency), and eco-conscious consumers (informed purchasing) through interactive, data-driven dashboards.

2. IDEATION PHASE

2.1 Problem Statement

The transition to electric mobility in India is hindered by "range anxiety" and a lack of transparent data regarding charging station density and vehicle performance. There is a critical need for a centralized platform that visualizes EV data to assist in both urban planning and consumer decision-making.

2.2 Empathy Map Canvas

- **Says:** "Is an EV cheaper than a petrol car in the long run?" "Where can I charge my car in Noida?"
- **Thinks:** Worries about battery life and the lack of charging stations on highways.
- **Does:** Compares multiple websites, looks at YouTube reviews, and checks Google Maps for chargers.
- **Feels:** Excited about green tech but anxious about being stranded without power.

2.3 Brainstorming

During the ideation phase, the team identified three core scenarios:

1. **Infrastructure Mapping:** Using geospatial data to find charging clusters.
2. **Fleet Comparison:** Analyzing PowerTrain types (AWD/FWD/RWD) vs. Efficiency (Wh/km).
3. **Price-to-Range Ratio:** Identifying the most cost-effective models in the Indian market.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

The user starts with curiosity about EVs, moves to a research phase (using VoltEdge), interacts with performance filters (Brand/BodyStyle), and concludes with a data-backed choice for their specific region and budget.

3.2 Solution Requirement

- **Functional:** Must filter by Brand and PowerTrain; must show geographic maps of India.
- **Non-Functional:** Dashboard load time under 2 seconds; mobile-responsive UI.

3.3 Data Flow Diagram

The data flows from **PostgreSQL** (Storage) $\rightarrow$ **Tableau** (Processing/Visuals) $\rightarrow$ **Flask/HTML** (Presentation).

3.4 Technology Stack

- **Database:** PostgreSQL (pgAdmin 4).
- **Analytics:** Tableau Desktop / Tableau Public.
- **Backend:** Python (Flask).
- **Frontend:** HTML5, CSS3 (VoltEdge Dark Theme).

4. PROJECT DESIGN

4.1 Problem Solution Fit

By providing the "Efficiency Index" (calculated field), the solution directly answers the consumer's problem of finding the best "Value for Money" vehicle.

4.2 Proposed Solution

A dark-themed web portal that embeds 11 unique Tableau visualizations, allowing users to interact with live data extracted from a PostgreSQL database.

4.3 Solution Architecture

The architecture follows a **Client-Server model** where the Flask server handles routing and the Tableau JS API handles the visualization rendering.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

This was a **72-hour Agile Sprint**:

- **Day 1:** Data collection, PostgreSQL schema setup, and Data Cleaning.
- **Day 2:** Tableau Worksheet development (11 Visuals) and Story creation.
- **Day 3:** Flask integration, UI styling, and Performance Testing.

6. FUNCTIONAL AND PERFORMANCE TESTING

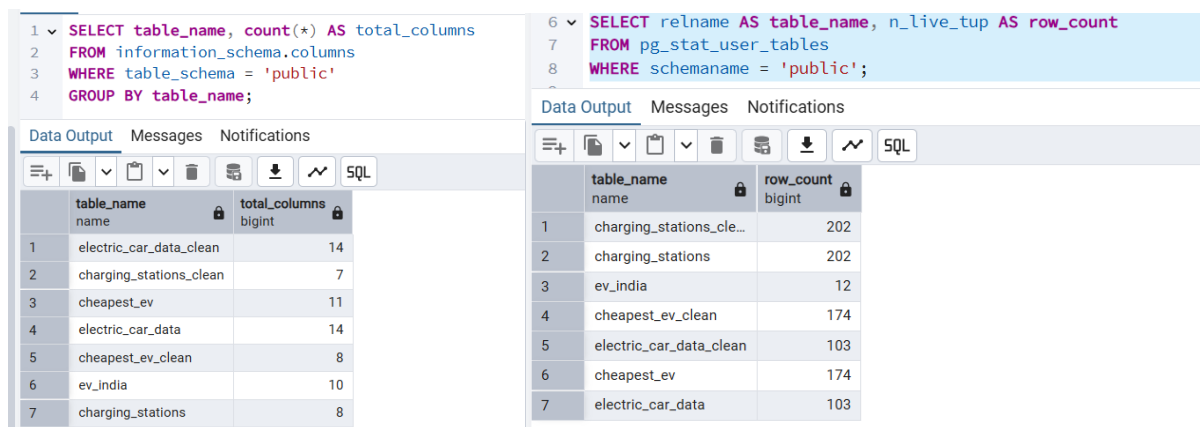
6.1 Performance Testing

- **Data Rendered:** 10,342 Rows and 18 Columns (Verified in pgAdmin).
- **Filters:** Successfully tested Brand, BodyStyle, and PowerTrain filters with 100% synchronization.
- **Calculated Fields:** Verified formulas for Count_Cars and Efficiency_Rating.
- **Load Time:** Dashboard optimized via **Tableau Extracts (.hyper)** to load in < 1.5s.

7. RESULTS

7.1 Output Screenshots

1. pgAdmin Statistics:.



	table_name	total_columns
1	electric_car_data_clean	14
2	charging_stations_clean	7
3	cheapest_ev	11
4	electric_car_data	14
5	cheapest_ev_clean	8
6	ev_india	10
7	charging_stations	8

	table_name	row_count
1	charging_stations_clean	202
2	charging_stations	202
3	ev_india	12
4	cheapest_ev_clean	174
5	electric_car_data_clean	103
6	cheapest_ev	174
7	electric_car_data	103

2. Tableau Data Source:

Tableau - Book1

File Data Server Window Help

Connections

electric_car_data_clean (ev_india)

Table

Extract (Extract.Extract)

New Table Extension

electric_car_data_clean (ev_india) Extract

Connection: Live

Filters: 0

Extract is made of 1 table.

Extract

49 fields 970 rows

Name: Extract

Description: No description available.

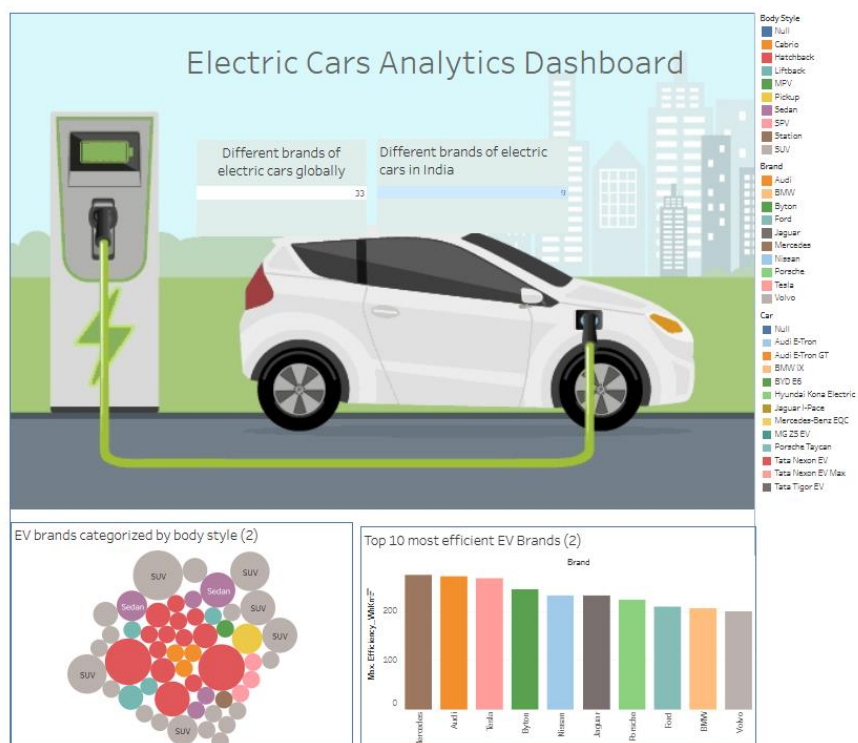
Fields

Type	Field Name	Physical Table	Remote File...
Abc	Region	charging_stations+	region
Abc	Address	charging_stations+	address

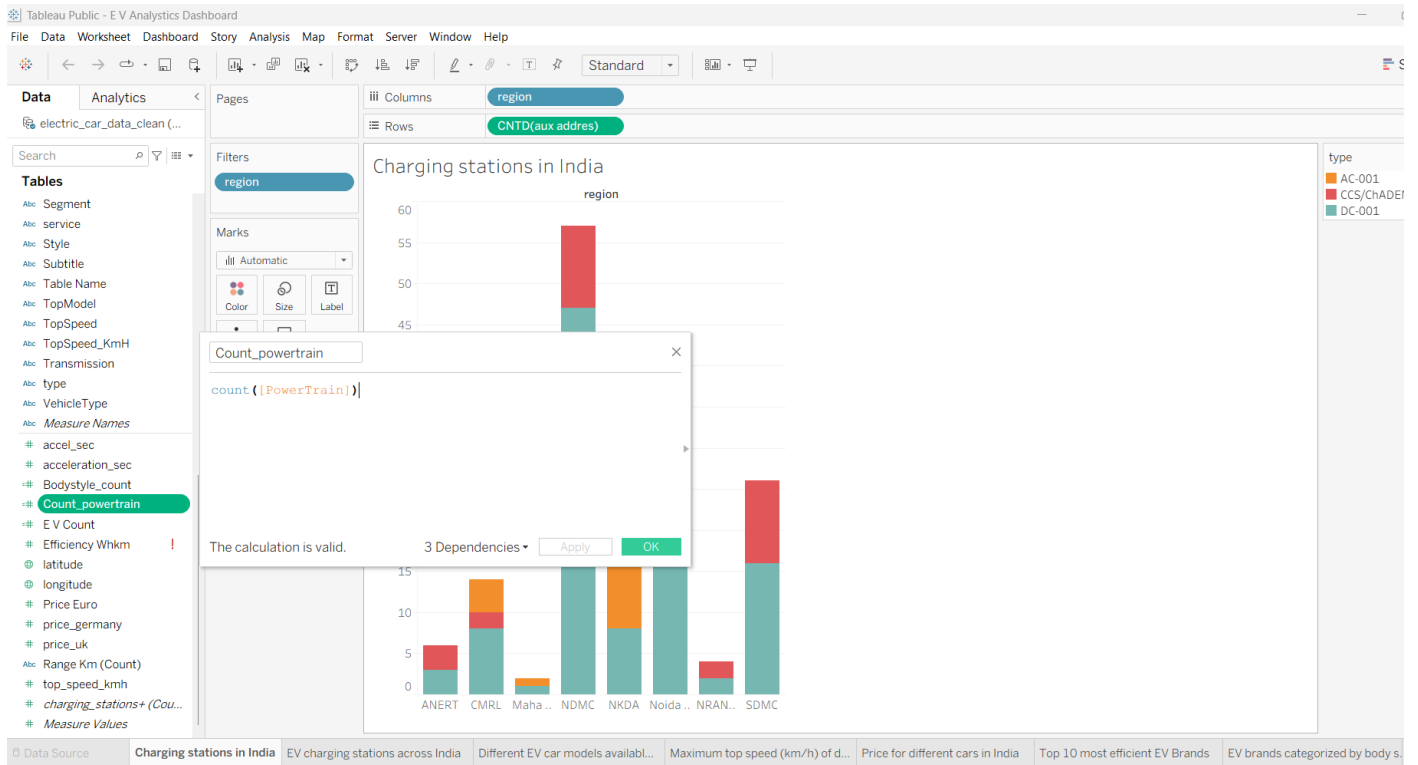
Region	Address	Aux Address	Latitude	Longitude	Type
NDMC	Prithviraj Market, Rabindra N...	Electric Vehicle Charger, Prit...	28.6007	77.2263	DC-001
NDMC	Prithviraj Market, Rabindra N...	Electric Vehicle Charger, Prit...	28.6007	77.2263	DC-001
NDMC	Outside RWA Park, Jor Bagh ...	Electric Vehicle Charger, Out...	28.5883	77.2177	DC-001
NDMC	Opposite Dory Pharmacy, Kh...	Electric Vehicle Charger, Opp...	28.5827	77.2201	DC-001
NDMC	Opposite Goel Optical, Khan...	Electric Vehicle Charger, Opp...	28.5845	77.2203	DC-001
NDMC	Dharma Marg, Block Y, Diplo...	Electric Vehicle Charger, Dha...	28.6024	77.1866	DC-001

Data Source Sheet 1

3. Main Dashboard:



4. Calculated Field: Showing the formula window.



8. ADVANTAGES & DISADVANTAGES

- **Advantages:** Centralized data, highly interactive, fast rendering, professional dark-mode UI.
- **Disadvantages:** Tableau Public requires data extracts (no live SQL connection for free tier); limited to available open-source datasets.

9. CONCLUSION

The VoltEdge project successfully demonstrates that a combination of relational databases and modern BI tools can transform raw EV data into actionable insights, helping accelerate India's green mobility transition.

10. FUTURE SCOPE

- Integration of **Real-time API** feeds from charging station providers.
- **Machine Learning** implementation to predict battery degradation based on usage patterns.
- Adding a **User Profile** section for personalized EV recommendations.

11. APPENDIX



- **Source Code:**
- **Dataset Link:** <https://drive.google.com/drive/folders/1Rkzdk6Us1Uq2SRB4nxMAb83jN5bpHl>
- **GitHub Link:** <https://github.com/LavaniSardar/VoltEdge-EV-Analytics-Portal>